

Lab Title

SOC Investigation. Firewall Alert for Blacklisted URL Access

Platform

TryHackMe SOC Simulation Lab

Objective

Investigate a firewall alert triggered by an internal host attempting to access a blacklisted external URL. Identify indicators of compromise, determine alert classification, and document response actions.

Alert Overview

Alert ID

8816

Alert Type

Access to Blacklisted External URL Blocked by Firewall

Severity

High

Data Source

Firewall Logs

Date and Time of Activity

March 9th 2026, 03:55

Alert Description

The firewall generated an alert when an internal system attempted to access a URL that exists on the organization's threat intelligence blacklist. The firewall blocked the connection attempt according to the "Blocked Websites" security rule.

Dashboard

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Created by

Alert queue

0 alerts incoming

Assigned alert(s)

Write case report

8816

Access to Blacklisted External URL Blocked by Firewall

High

Firewall

Mar 9th 2026 at 03:57

Description:

This alert was triggered when a user attempted to access an external URL that is listed in the organization's blacklist or threat intelligence feeds. The firewall or proxy successfully blocked the outbound request, preventing the connection. Note: The blacklist only covers known threats. It does not guarantee protection against new or unknown malicious domains.

datasource:

firewall

timestamp:

03/09/2026 03:55:41.836

Action:

blocked

SourceIP:

10.20.2.17

SourcePort:

34257

DestinationIP:

67.199.248.11

DestinationPort:

80

URL:

http://bit.ly/3sHk3da12340

Application:

web-browsing

Protocol:

TCP

Rule:

Blocked Websites

Search for an alert

Reset filters

Severity

Status

Alert type

Show 15 alerts

| ID   | Alert rule                                        | Severity | Type     | Date                  | Status          | Action |
|------|---------------------------------------------------|----------|----------|-----------------------|-----------------|--------|
| 8818 | Inbound Email Containing Suspicious External Link | Medium   | Phishing | Mar 9th 2026 at 03:59 | Awaiting action |        |

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You haven't picked up any alert! Assign yourself to an alert to start investigating and find all the true positives. [Learn more](#)

Search for an alert

Reset filters

Severity

Status

Alert type

Show 15 alerts

| ID                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Alert rule                                             | Severity | Type     | Date                  | Status          | Action |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|----------|----------|-----------------------|-----------------|--------|
| 8817                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Inbound Email Containing Suspicious External Link      | Medium   | Phishing | Mar 9th 2026 at 03:58 | Awaiting action |        |
| 8816                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Access to Blacklisted External URL Blocked by Firewall | High     | Firewall | Mar 9th 2026 at 03:57 | Awaiting action |        |
| <div><div>Description:</div><div>This alert was triggered when a user attempted to access an external URL that is listed in the organization's blacklist or threat intelligence feeds. The firewall or proxy successfully blocked the outbound request, preventing the connection. Note: The blacklist only covers known threats. It does not guarantee protection against new or unknown malicious domains.</div><div><div>datasource:</div><div>firewall</div><div>timestamp:</div><div>03/09/2026 03:55:41.836</div><div>Action:</div><div>blocked</div><div>SourceIP:</div><div>10.20.2.17</div><div>SourcePort:</div><div>34257</div><div>DestinationIP:</div><div>67.199.248.11</div><div>DestinationPort:</div><div>80</div><div>URL:</div><div>http://bit.ly/3sHk3da12340</div><div>Application:</div><div>web-browsing</div><div>Protocol:</div><div>TCP</div><div>Rule:</div><div>Blocked Websites</div></div></div> |                                                        |          |          |                       |                 |        |
| 8815                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Inbound Email Containing Suspicious External Link      | Medium   | Phishing | Mar 9th 2026 at 03:56 | Awaiting action |        |
| 8814                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Inbound Email Containing Suspicious External Link      | Medium   | Phishing | Mar 9th 2026 at 03:53 | Awaiting action |        |

Technical Details from Alert

Action

Blocked

Source IP Address

10.20.2.17

Source Port

34257

Destination IP Address

67.199.248.11

Destination Port

80

Protocol

TCP

URL Requested

<http://bit.ly/3sHkX3da12340>

Application

Web Browsing

Firewall Rule Triggered

Blocked Websites

## Investigation Steps

### Step 1. Review Alert Information

The alert indicates an internal host attempted to access a blacklisted external URL. The firewall blocked the request, preventing communication with the external destination.

### Observation

The URL uses a shortened link format through the bit.ly service. URL shortening services often conceal malicious destinations and appear frequently in phishing campaigns.

### Step 2. Identify Indicators of Compromise

Indicators discovered during the investigation.

Suspicious shortened URL

<http://bit.ly/3sHkX3da12340>

External destination IP

67.199.248.11

Internal source IP

10.20.2.17

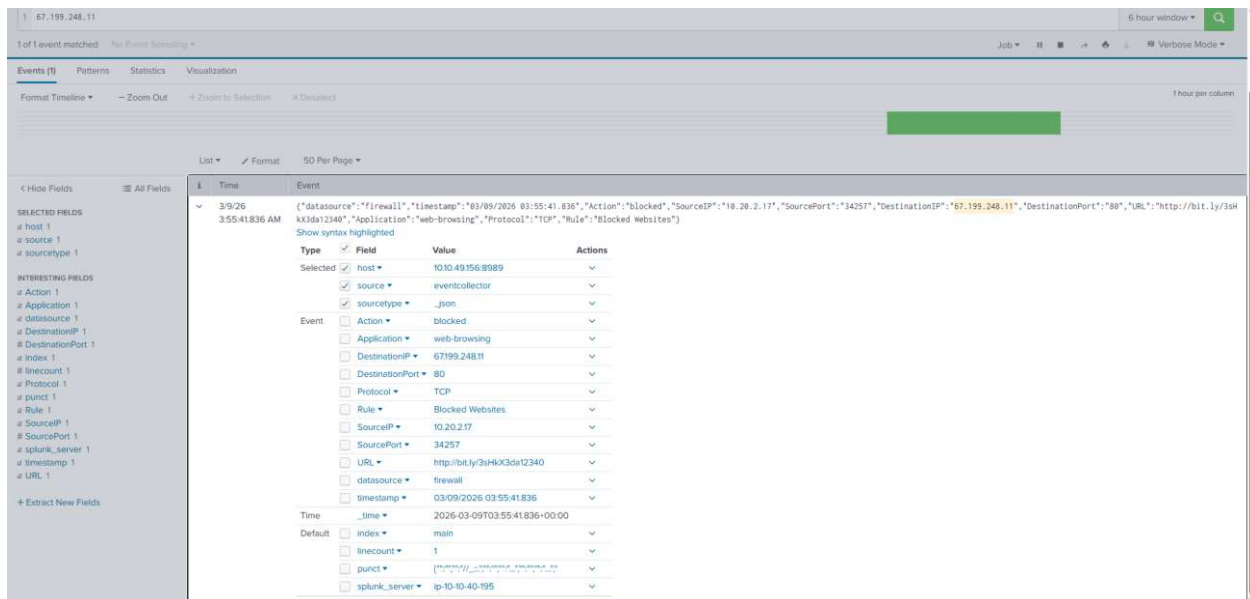
Security rule triggered

Blocked Websites

The screenshot displays the Splunk Enterprise web interface. At the top, the navigation bar includes 'splunk-enterprise', 'Apps', 'Messages', 'Settings', 'Activity', 'Help', and a search bar. Below this, the 'Search' tab is active, showing a 'New Search' bar with the query 'index=main 7.199.248.11'. The search results show '1 of 1 event matched' with 'No Event Sampling'. The event is displayed in a table with columns 'Time' and 'Event'. The event details are as follows:

| Time                  | Event                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3/9/26 3:55:41.836 AM | <pre>{   "Action": "blocked",   "Application": "web-browsing",   "DestinationIP": "67.199.248.11",   "DestinationPort": 80,   "Protocol": "TCP",   "Rule": "Blocked Websites",   "SourceIP": "10.20.2.17",   "SourcePort": 34257,   "URL": "http://bit.ly/3sHkX3da12340",   "datasource": "firewall",   "timestamp": "03/09/2026 03:55:41.836" }</pre> <p>Show as raw text<br/>host = 10.10.49.156.8989   source = eventcollector   sourcetype = _json</p> |

The interface also shows a sidebar with 'SELECTED FIELDS' (host, source, sourcetype) and 'INTERESTING FIELDS' (Action, Application, datasource, DestinationIP, index, linecount, Protocol, punct, Rule, SourceIP, SourcePort, splunk\_server, timestamp, URL). At the bottom, there is a link to 'Extract New Fields'.



## SIEM Query Used

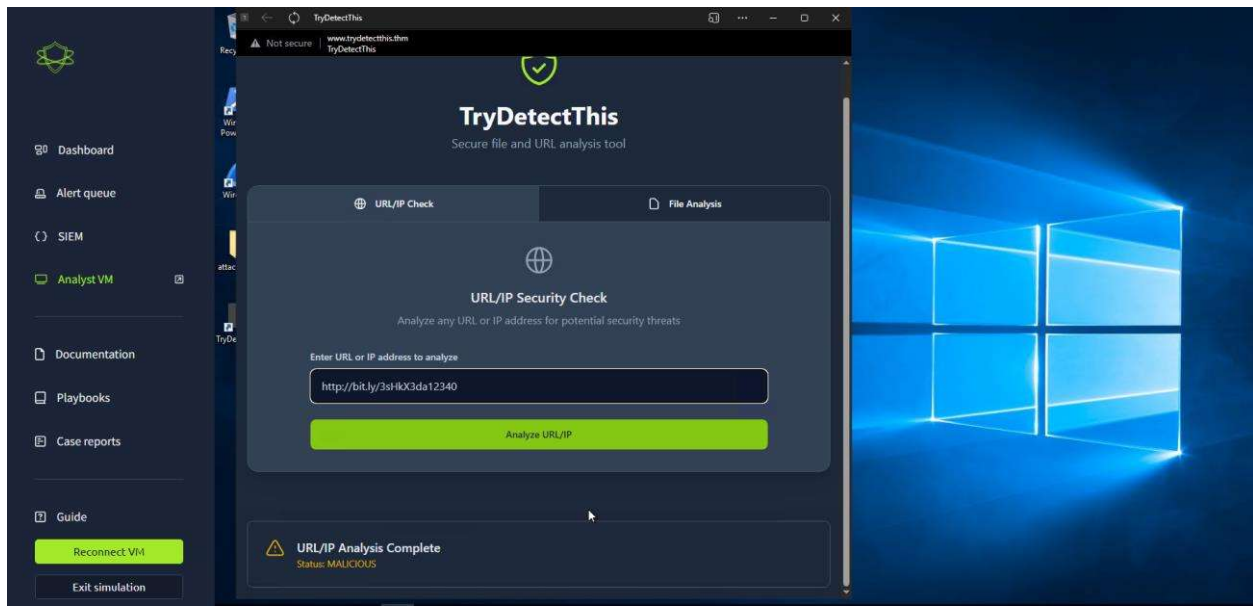
index=main 67.199.248.11

## Purpose of Query

The query searched firewall logs for activity involving the suspicious destination IP address observed in the alert.

## Step 3. Correlate with Other Security Alerts

The investigation checks whether the shortened URL appears in other alerts such as phishing emails or web proxy logs. This step determines whether the user accessed the link through a phishing message.



#### Step 4. Assess Firewall Action

The firewall security control successfully blocked the outbound request. The endpoint did not establish communication with the malicious destination.

#### Observation

Although the connection was blocked, the alert confirms that an internal user or process attempted to access a malicious URL.

#### Alert Classification

#### True Positive

#### Reason

The internal system attempted to access a URL already identified as malicious and listed on the organization's blacklist. The firewall blocked the request, preventing the connection from being established.

#### Escalation Decision

#### Escalation Required

## Reason

The attempt indicates potential phishing activity or user interaction with malicious content. Escalation allows the security team to verify whether the endpoint is compromised and whether additional users attempted to access the same malicious link.

## Recommended Remediation Actions

Identify the user associated with internal IP address 10.20.2.17.

Notify the user about the attempted malicious access.

Search email systems for phishing messages containing the shortened URL.

Confirm that the malicious domain remains blocked across firewall and proxy controls.

Monitor the endpoint for additional suspicious activity.

The screenshot displays a SIEM dashboard with a dark theme. On the left is a sidebar with navigation links: Dashboard, Alert queue, SIEM, Analyst VM, Documentation, Playbooks, Case reports, and Guide. The main area is titled 'Assigned alert(s)' and shows a table with one alert. A modal window is open over the alert, asking 'Was this alert a true positive or a false positive?' with 'True positive' selected. Below the modal is a table of alerts.

| ID   | Alert rule                                             | Severity | Type     | Date                  | Status          | Action |
|------|--------------------------------------------------------|----------|----------|-----------------------|-----------------|--------|
| 8816 | Access to Blacklisted External URL Blocked by Firewall | High     | Firewall | Mar 9th 2026 at 03:57 |                 |        |
| 8818 | Inbound Email Containing Suspicious External Link      | Medium   | Phishing | Mar 9th 2026 at 03:59 | Awaiting action |        |
| 8817 | Inbound Email Containing Suspicious External Link      | Medium   | Phishing | Mar 9th 2026 at 03:58 | Awaiting action |        |
| 8815 | Inbound Email Containing Suspicious External Link      | Medium   | Phishing | Mar 9th 2026 at 03:56 | Awaiting action |        |

## Final Outcome

The investigation confirmed that an internal endpoint attempted to access a blacklisted malicious URL. The firewall security control blocked the connection, preventing communication with the external host. The alert was escalated for further monitoring and investigation of the associated endpoint.



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0 alerts incoming

Assigned alert

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Search for an alert

Reset filters

Severity

Closed

Alert type

Show

15

alerts

| ID   | Alert rule                                             | Severity | Type     | Date                  | Status | Action |
|------|--------------------------------------------------------|----------|----------|-----------------------|--------|--------|
| 8816 | Access to Blacklisted External URL Blocked by Firewall | High     | Firewall | Mar 9th 2026 at 03:57 | Closed |        |